Project Melampo

A Technical, Scientific, Medical and Physiological Treatise on Multimodal Computer-Aided Intuition

Author: Francesco Lattari

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Generated from the Melampo GitHub documentation corpus and publicly accessible scientific sources.

Clinical safety statement: Project Melampo is a research scaffold. It is not a validated medical device, does not provide autonomous clinical diagnosis, and must not be used for patient-care decisions without formal clinical validation, regulatory review, monitoring, and human specialist oversight.

Source acquisition note: Automated retrieval of the public website melampo.chandra.it timed out during preparation. This edition therefore does not assert site-specific technical claims beyond the intended source locus; the treatise is grounded primarily in the current GitHub documentation corpus and publicly accessible supporting scientific and technical sources.

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1. Abstract and thesis

Project Melampo is a research architecture for Computer-Aided Intuition in medicine. Its central claim is that a diagnostic system should not be reduced to a single model, a single vector store, or a single language interface. It should instead be organized as a governed multimodal organism: specialist models produce signals, functional areas contextualize them, a semantic memory graph grounds them, neuro-dynamic metrics estimate coherence and mismatch, an intuition engine proposes candidates, a differential engine tests alternatives, policy layers decide whether to abstain or escalate, and a final diagnostic orchestrator emits a structured research result.

The architecture is intentionally multimodal and multimodel. Radiology and volumetric imaging are assigned to a Pillar-0 oriented signal pathway; grounded clinical text and agentic reasoning are assigned to a Gemma 4 oriented pathway; external scientific critique can be delegated to a Claude Healthcare or Life Sciences style critic; documents are parsed through a Docling-oriented ingestion contract; semantic memory is designed around a Weaviate object-property graph; the final authority belongs to MelampoDiagnosticOrchestrator rather than any external model. This separation is not cosmetic. It is the safety condition that prevents linguistic fluency, image confidence, or retrieval similarity from being mistaken for clinical truth.

The physiological inspiration of Melampo does not assert that software modules literally reproduce the brain. Instead, it uses selected principles from neuroscience and clinical cognition as design constraints: prediction error, inhibitory control, salience, convergence, memory replay, contextual belief updating, mismatch resolution, and controlled abstraction. In this treatise those principles are translated into computational contracts, metrics, and governance requirements. The result is a research scaffold for building auditable diagnostic reasoning systems, not a deployed medical device.

The principal thesis

A clinically meaningful artificial intuition cannot be generated by ungrounded generation alone. It must emerge from the controlled interaction among perception, semantic memory, context, uncertainty, contradiction, and policy. This is the core architectural thesis of Melampo.

Figure 1. Melampo high-level multimodal and multimodel architecture

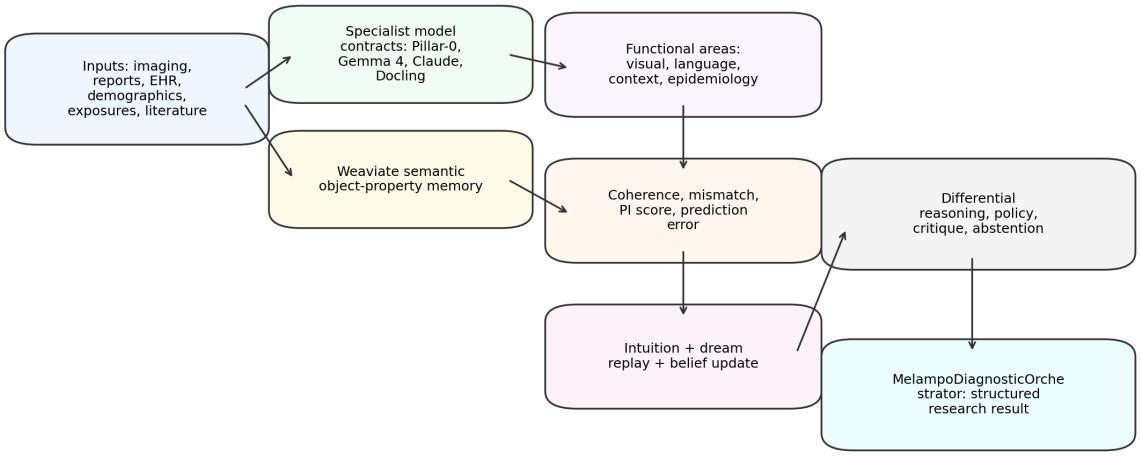

Figure 1. A high-level view of the Melampo architecture: specialist signal providers feed functional areas; functional areas feed neuro-dynamic metrics and intuition; the final structured research result is produced by the Melampo-owned orchestrator.

2. Source corpus, method and clinical boundary

This treatise is synthesized from the current Melampo GitHub documentation corpus, including the final treatise decision record, the current architecture document, the enterprise AI/RAG evolution plan, the validation framework, the model stack card, and the dataset card. These internal sources define the project as a multimodal, multimodel, neuro-symbolic and neuro-inspired research framework designed to evolve Computer-Aided Diagnosis into Computer-Aided Intuition [M1-M5].

Public technical and scientific sources are used to contextualize model and infrastructure choices. Weaviate documentation establishes the relevance of data objects, properties, cross-references, and named vectors to object-property semantic memory [S1-S3]. Docling documentation establishes a document ingestion pathway for PDF, DOCX, PPTX, XLSX, HTML, images, audio, OCR, tables, formulas, and structured export formats for RAG and agentic systems [S4-S6]. Pillar-0 sources motivate the replacement of a general medical multimodal model with a radiology foundation model for volumetric CT/MRI interpretation [S7-S8]. Anthropic sources motivate the optional external critic role in life sciences and healthcare workflows rather than final diagnostic authority [S9-S10]. Neuroscience and calibration sources ground the predictive-processing, active-inference, free-energy, calibration, and clinical evaluation language used in the design [S11-S16].

The public site melampo.chandra.it was requested during preparation. Automated retrieval timed out. The absence of retrievable content prevents direct incorporation of site-specific claims in this edition. A future revision should re-ingest that site, extract its canonical language, and reconcile it against the GitHub documentation corpus.

Clinical boundary

- Melampo is not a validated medical device.
- Melampo outputs are research outputs, not patient-care decisions.
- The architecture requires human specialist oversight before any clinical interpretation.
- Prospective clinical validation, regulatory classification, monitoring, and governance are prerequisites for deployment.
- External models are not final arbiters. They are signal providers, critics, parsers, retrievers, or explainers.

3. Foundational identity of Project Melampo

The foundational identity of Melampo is explicit: it is not a single-model diagnostic assistant and not a simple retrieval-augmented generation wrapper. It is a multimodal, multimodel, neuro-symbolic and neuro-inspired research framework. The GitHub decision record states the canonical sequence as specialist models feeding functional areas, functional areas feeding coherence and mismatch dynamics, coherence and mismatch feeding intuition, intuition feeding differential reasoning, and differential reasoning feeding policy, critique, governed memory, and dream replay [M1].

This identity has several consequences. First, no specialist model can be allowed to collapse the full diagnostic state by itself. A radiology model may detect or prioritize visual findings; a language model may summarize and reason over text; a critic may identify contradictions; a document parser may extract evidence; a vector database may retrieve semantically similar cases. None of these components is sufficient to claim a diagnosis. The final research output requires a system-level controller that integrates their signals under uncertainty.

Second, the architecture is designed to preserve contextual richness. In medicine, data are dangerous when detached from context. A symptom without its temporal profile, a radiological pattern without anatomical localization, a guideline excerpt without publication provenance, or a model score without calibration can all mislead. Melampo therefore privileges object-property memory, provenance, relations, uncertainty, and explicit abstention.

Concept	Melampo interpretation	Architectural consequence
Computer-Aided Diagnosis	Machine support for candidate identification and evidence retrieval.	Useful but insufficient when reduced to classification.
Computer-Aided Intuition	System-level emergence of diagnostic candidates from convergence, memory, mismatch, and controlled belief update.	Requires functional areas, semantic memory, dream replay, metrics, policy, and audit.
External models	Specialist signal providers or critics.	Never sole final diagnostic arbiters.
Semantic memory	Object-property clinical graph with vectors and relations.	Weaviate is preferred over vector-only memory.
Validation	Retrospective benchmark, calibration, prospective prediction locking, and clinical governance.	Performance claims require dataset cards and prospective evaluation.

4. Physiological and neurocomputational foundations

4.1 The brain as an inference system

Melampo draws inspiration from predictive processing, predictive coding, active inference, and free-energy based accounts of perception and action. In these frameworks, biological agents are not passive receptors of sensory data. They continuously generate predictions, compare those predictions with incoming sensory evidence, and update internal states when mismatch or prediction error becomes significant. Recent reviews emphasize both the explanatory promise and the need for empirical caution: predictive coding and active inference remain powerful frameworks, but their empirical status is not uniformly settled across all domains [S11-S12].

The computational translation for Melampo is therefore conservative. The system does not claim to implement a biological cortex. Instead, it uses predictive processing as a design metaphor and formal constraint: each functional area emits signals with salience and uncertainty; cross-area interactions produce coherence and mismatch measures; unresolved mismatch creates revision pressure; high precision and high convergence support candidate stabilization; uncertainty and conflict trigger abstention, escalation, or dream branch exploration.

The free-energy principle suggests that adaptive systems act to reduce surprise through perception, learning, and action. In Melampo, this translates into a controlled reduction of diagnostic surprise. A candidate is not accepted because it is fluent or salient; it is accepted provisionally only when supported by multimodal convergence, semantic grounding, low contradiction, sufficient calibration, and policy permission. This makes the analogy operational rather than metaphysical.

4.2 Action potentials, gain and signal reliability

At the cellular level, biological neurons communicate through electrochemical events, including graded synaptic potentials and all-or-none action potentials. The relevance to Melampo is not an attempt to simulate ion channels; the relevance is the separation between signal generation, signal propagation, signal gain, and downstream interpretation. A strong signal is not automatically a correct signal. Biological systems regulate gain through attention, neuromodulation, inhibition, recurrent context, and homeostatic constraints. Melampo mirrors this separation by making model outputs only one element of an area signal and by requiring confidence, uncertainty, provenance, and limitations.

The design distinction between salience and validity is essential. A radiological model can produce a high-salience finding; a language model can produce a plausible narrative; a semantic memory query can retrieve a closely related case. In isolation, each signal may be misleading. The neuro-dynamic layer therefore treats signal strength as only one component. It computes prediction error, mismatch ratio, convergence index, inhibitory control, deductive gate, candidate temperature, and belief update rate. These metrics formalize the idea that signal gain must be regulated by context and contradiction.

4.3 Cortical hierarchy, prediction error and clinical mismatch

Predictive coding theories describe hierarchical exchange between top-down predictions and bottom-up

prediction errors. The clinical analogue is differential diagnosis. A candidate pathology predicts symptoms, imaging findings, epidemiological factors, laboratory abnormalities, chronology, and response patterns. When the evidence fails to match those predictions, the hypothesis must be revised. Melampo implements this principle as an explicit computational loop: candidates generate expectations, evidence produces support and contradiction, and the mismatch index determines whether the system stabilizes, revises, dreams, escalates, or abstains.

This approach is particularly important in medicine because diagnostic expertise is not merely pattern recognition. Expert clinicians detect when a case does not fit. They notice discordance between a radiograph and a narrative, between epidemiology and presentation, between expected progression and actual time course, between one organ system and another. Melampo encodes this property through cross-area pair profiles and neuro-dynamic metrics. The clinical value lies not only in recognizing pneumonia, hemorrhage, tumor, or infection, but in recognizing when the present case is not sufficiently explained by the leading candidate.

Figure 2. Neuro-dynamic integration of functional areas

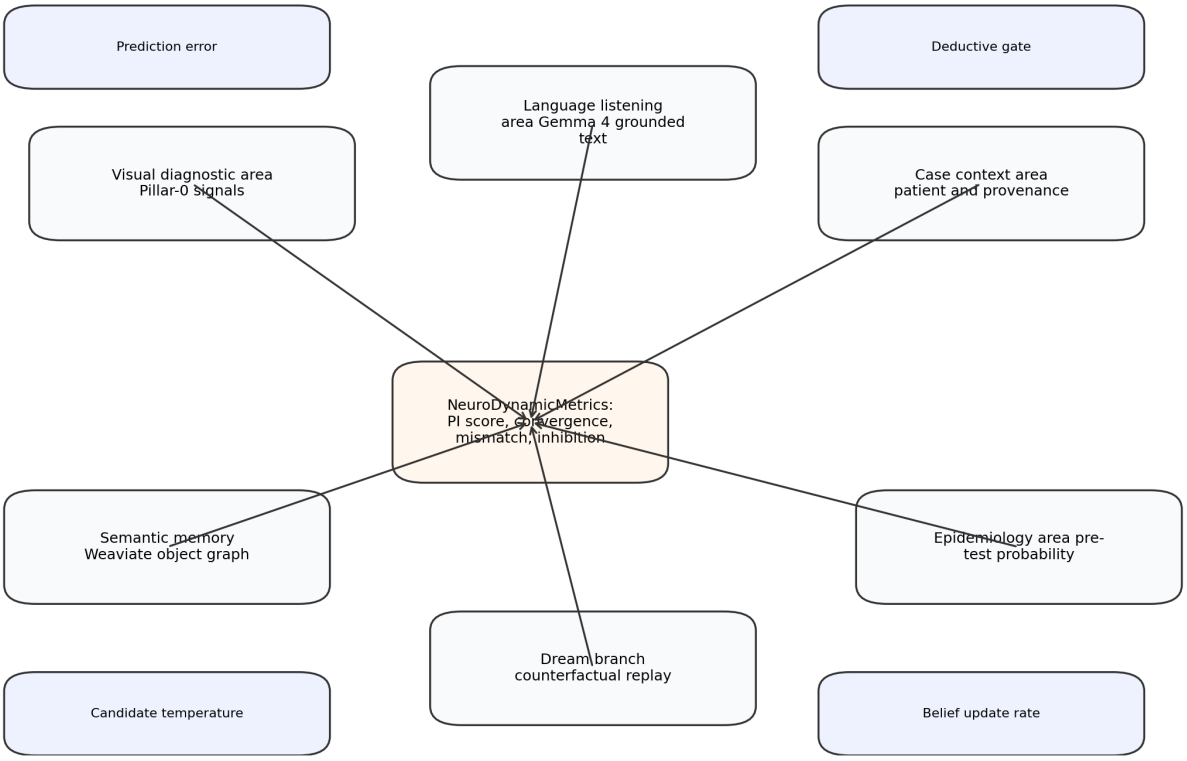

Figure 2. Melampo interprets functional-area signals through a neuro-dynamic layer that estimates coherence, prediction error, mismatch, inhibition, candidate temperature, dream plasticity, and belief update rate.

5. Medical reasoning and the physiology of diagnostic intuition

Medical intuition is often misunderstood as an unexplainable shortcut. In expert practice, however, intuition is typically a compressed result of repeated exposure, memory, pattern abstraction, error correction, and contextual reasoning. It can be fast without being arbitrary. It can be tacit without being irrational. It can also be biased, overconfident, and unsafe when not checked by deliberate reasoning. Melampo treats intuition as a candidate generation state that must be inspected, challenged, calibrated, and governed.

The physiological analogy is useful. The brain integrates sensory streams, bodily states, memory traces, attention, salience, reward, affect, and action readiness. The diagnostic mind integrates visual findings, patient narrative, epidemiology, prior disease knowledge, contextual constraints, and risk. A system that aspires to

computer-aided intuition must therefore integrate across modalities and must preserve the ability to reject its own first impression.

Melampo defines intuition as an emergent diagnostic candidate produced by multimodal convergence, semantic memory, coherence between simulated functional areas, controlled mismatch resolution, inhibition of noisy or bias-prone signals, dream replay and contextual belief update [M1]. This definition rejects both black-box mysticism and naive classification. It places intuition inside a structured architecture that can be audited.

5.1 Differential diagnosis as controlled hypothesis competition

Differential diagnosis is a competition among hypotheses under incomplete evidence. It is not merely a ranked list. Each hypothesis has expected features, missing features, contradictory features, prevalence context, severity, treatability, and risk of delay. Melampo therefore couples the intuition engine to the differential engine and policy stack. A high-scoring candidate is converted into a structured differential state only after evidence support, contradiction, uncertainty, and safety gates are evaluated.

The architecture requires that every leading hypothesis be accompanied by supporting evidence, contradicting evidence, uncertainty, missing evidence, and recommended actions. This resembles expert clinical reasoning, where the question is not only what disease is likely but also what dangerous alternative must be excluded, what evidence is insufficient, what test would change management, and when a specialist must intervene.

6. Enterprise multimodal architecture

The current architecture document describes Melampo as a system combining specialist model signals, simulated clinical functional areas, semantic memory, neuro-dynamic metrics, dream/replay, differential reasoning, policy control, and an audit-first diagnostic orchestrator [M2]. This structure is enterprise-oriented because it separates contracts, providers, audit, validation, and final authority. Each external model is replaceable; each output must be structured; each live integration must be explicit; each clinical claim must be traceable.

Enterprise-grade does not mean production-ready clinical deployment. It means the system is organized according to constraints that make future production hardening possible: typed public contracts, deterministic fallback behavior, provider neutrality, no hidden network calls, provenance, audit metadata, explicit limitations, tests for normal and adverse paths, and no automatic promotion of synthetic material into clinical truth [M3].

Layer	Purpose	Current implementation status
Input layer	Payloads for imaging, reports, EHR text, patient narrative, demographics, exposures, documents.	Prototype payloads and dataset loaders are scaffolded.
Specialist signal layer	Pillar-0, Gemma 4, Claude-style critic, Docling, Weaviate contracts.	Contracts are implemented; live inference requires infrastructure subclasses.
Functional area layer	Visual, language, context, epidemiology areas.	Area modules emit structured salience and signal metadata.
Neuro-dynamic layer	PI score, prediction error, convergence, mismatch, inhibition, belief update.	Metric scaffold implemented; requires calibration.
Reasoning layer	Intuition, dream replay, differential reasoning, critique.	Scaffold implemented with heuristic scoring.
Final controller	MelampoDiagnosticOrchestrator.	Implemented as final research authority.
Validation layer	Benchmark, prospective registry, calibration.	Primitives implemented; no real datasets bundled.

7. Specialist model strategy

7.1 Pillar-0 for radiology and volumetric imaging

The model strategy designates Pillar-0 as the primary radiology and volumetric imaging foundation model candidate. The rationale is clinical and architectural. Melampo is strongly oriented toward CT/MRI volumetric

reasoning. Pillar-0 sources describe a radiology foundation model pretrained on large CT and MRI corpora and evaluated across hundreds of radiologic findings [S7-S8]. Its role in Melampo is not to produce the final diagnosis. Its role is to produce visual diagnostic signals: suspected findings, anatomical localization, confidence, uncertainty, and mismatch cues for the visual diagnostic area.

The substitution of MedGemma by Pillar-0 is therefore partial and role-specific. It is not an elimination of medical language reasoning. Pillar-0 should occupy the imaging foundation pathway because volumetric radiology requires attention to grayscale contrast, anatomy, 3D context, and clinically grounded imaging findings. The final output remains controlled by MelampoDiagnosticOrchestrator.

7.2 Gemma 4 for grounded clinical language and agentic reasoning

Gemma 4 is designated for non-radiology language functions: report summarization, EHR text interpretation, symptom chronology, grounded reasoning, and local or private agentic workflows. Public reporting describes Gemma 4 as an open-weight multimodal model family with Apache 2.0 licensing and long-context capabilities; the treatise treats it as a candidate that requires local verification, model card review, deployment governance, and RAG grounding before any clinical role [S17-S18].

In Melampo, Gemma 4 is not a clinical oracle. It is a language and workflow signal provider. Its outputs must be bound to semantic memory, source provenance, confidence, uncertainty, and limitations. A plausible textual explanation without evidentiary grounding is insufficient.

7.3 Claude-style external critic

Anthropic describes Claude for Life Sciences and healthcare-related capabilities as supporting scientific research, clinical operations, regulatory workflows, and connectors to scientific platforms [S9-S10]. In Melampo this maps naturally to an optional external critic. Such a critic can review hypotheses, check literature synthesis, identify regulatory or compliance concerns, and evaluate report quality. It must not become the final diagnostic authority.

The critic role is deliberately external and optional. This preserves sovereignty of the Melampo reasoning core, avoids hidden dependency on a proprietary service, and enables environments where data policy prevents external transmission.

Model or system	Role in Melampo	Prohibited role
Pillar-0	Radiology and volumetric imaging signal provider.	Final diagnostic arbiter.
Gemma 4	Grounded clinical text and workflow reasoning provider.	Ungrounded medical oracle.
Claude Healthcare / Life Sciences style critic	External second opinion, literature/regulatory critic.	Primary imaging engine or final arbiter.
Docling	Document parser and chunk preparation tool.	Reasoning model.
Weaviate	Semantic object-property memory and ontology RAG.	Reasoning model or clinical judge.
MelampoDiagnosticOrchestrator	Final internal research controller.	Autonomous medical device without validation.

8. Semantic memory, ontology and RAG

The enterprise plan selects Weaviate as the primary semantic memory backend because Melampo prioritizes semantic knowledge, object-property relations, ontology-like clinical structure, and multimodal context over raw vector throughput alone [M3]. Weaviate documentation describes data objects stored in collections with properties, vectors, cross-references, and named vectors [S1-S3]. This maps directly to clinical reasoning: a symptom can suggest several pathologies; a pathology can have symptoms, imaging patterns, risk factors, and supporting documents; a case can link demographics, reports, images, differential hypotheses, and outcomes.

This memory model is intentionally more than vector similarity. In medicine, similarity without context can be dangerous. A retrieved case may be semantically close but clinically irrelevant because age, immune status,

geography, imaging modality, disease stage, or treatment context differs. Therefore, Melampo requires not only embeddings but also properties, references, ontology identifiers, provenance, learning status, and validation status.

Figure 3. Target Weaviate-style semantic clinical object graph

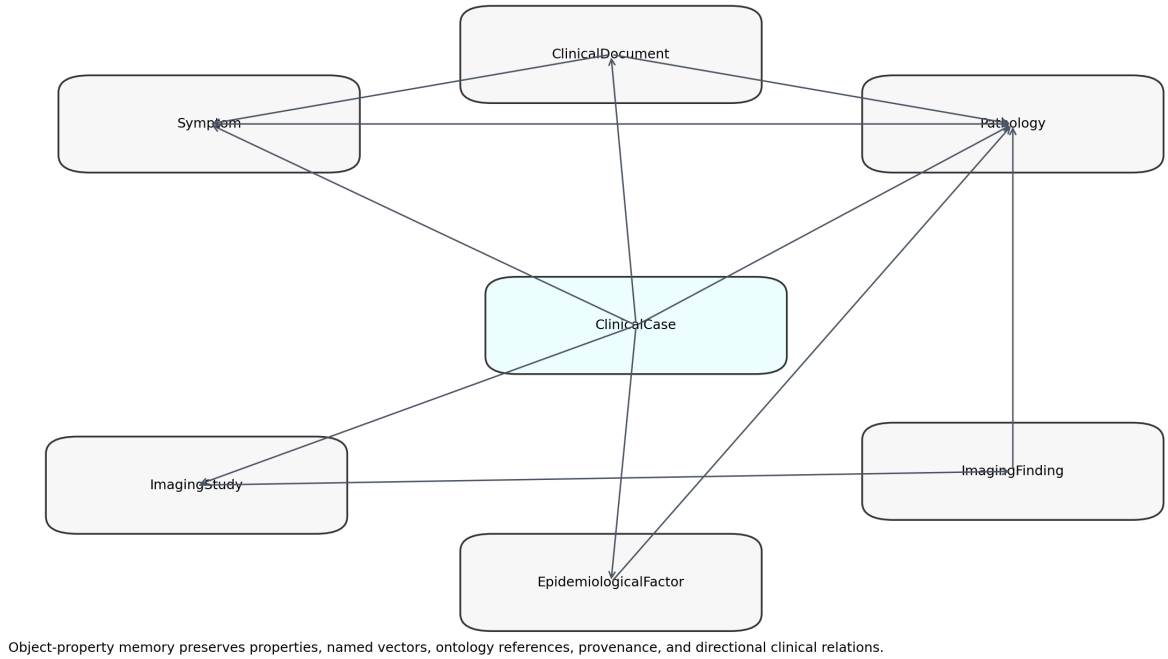

Figure 3. Target semantic clinical object graph. Weaviate is used conceptually as object-property memory with named vectors, references, and provenance, not as a vector-only store.

8.1 Target object classes

Object class	Clinical meaning	Examples of relations
Symptom	A normalized clinical complaint or observation.	suggestsPathology, appearsInCase.
Pathology	A candidate disease or pathological process.	hasSymptom, hasImagingPattern, hasRiskFactor.
ClinicalCase	A governed case memory object.	hasSymptom, hasReport, hasImage, hasDifferential.
ImagingStudy	An imaging object with modality and vectors.	belongsToCase, hasFinding.
ImagingFinding	A radiological or visual finding.	supportsPathology, foundInStudy.
ClinicalDocument	A guideline, report, article, or document chunk.	mentionsSymptom, mentionsPathology, mentionsFinding.
EpidemiologicalFactor	A risk factor, exposure, region, or prevalence variable.	increasesRiskOf, appearsInCase.

9. Document intelligence and evidence ingestion

Melampo requires a document ingestion layer because clinical reasoning depends on reports, guidelines, articles, tables, laboratory narratives, epidemiological sources, and local protocols. The enterprise plan recommends Docling as the production parser for clinical document ingestion [M3]. Public Docling documentation describes

parsing of multiple formats, advanced PDF understanding, layout, reading order, table structure, formulas, OCR, unified representation, and exports to Markdown, HTML, JSON, and other formats [S4-S6].

The key requirement is not simply text extraction. The parser must preserve provenance and structure. A table from a guideline, a figure caption from a radiology article, a section heading in a protocol, and a publication date all change the epistemic status of a retrieved statement. The output of document intelligence should therefore be a governed ClinicalDocument object, chunked by semantic section, linked to symptoms and pathologies, and inserted into semantic memory with source and license metadata.

Operational document ingestion sequence

1. Register the source catalog entry: URI, title, source type, license, publication date, version, and access class.
2. Convert with Docling when available; otherwise use plain-text fallback only for low-risk development material.
3. Preserve page, section, table, figure, formula, and reading-order metadata where available.
4. Chunk by clinical section and semantic boundary, not only fixed token count.
5. Normalize ontology mentions: symptoms, pathologies, findings, drugs, anatomical regions, and epidemiological factors.
6. Create ClinicalDocument objects and references to Symptom, Pathology, ImagingFinding, and EpidemiologicalFactor objects.
7. Insert into Weaviate with learning_status set to candidate unless curated and validated.
8. Expose retrieved evidence only with provenance, source type, and date.

10. Neuro-dynamic metrics and belief update

The neuro-dynamic layer is the computational bridge between functional-area signals and diagnostic intuition. It computes the predictive-inference quality score, precision-weighted coherence, prediction error, cross-area synchrony, mismatch ratio, signal precision, prior precision, conflict load, convergence index, mismatch index, inhibitory control, deductive gate, revision pressure, dream plasticity, intuition gain, candidate temperature, and belief update rate [M3].

These metrics are not presented as validated biomarkers. They are computational abstractions inspired by predictive processing, active inference, cross-area synchrony, inhibitory control, and recurrent integration. Their immediate role is to make candidate formation inspectable. A diagnostic candidate should reveal whether it emerged from high coherence, high mismatch, high dream pressure, high uncertainty, or high inhibition.

The candidate temperature is especially important. In low-conflict high-precision states, candidate temperature can fall, allowing exploitation of a stable hypothesis. In high-conflict states, candidate temperature rises, encouraging exploration, re-ranking, contradiction review, and dream branch generation. This formalizes the principle that uncertainty should widen reasoning rather than prematurely collapse it.

Metric	Physiological analogy	Computational role
pi_score	Predictive inference adequacy.	Composite quality signal for candidate plausibility.
prediction_error	Bottom-up mismatch.	Signals unresolved evidence contradicting predictions.
convergence_index	Cross-area synchrony.	Measures multimodal agreement.
mismatch_index	Conflict between predictions and signals.	Triggers revision, critique, or abstention.
inhibitory_control	Suppression of noisy /bias-prone activity.	Reduces overconfident candidates.
deductive_gate	Slow controlled reasoning.	Regulates transition from intuition to differential reasoning.
dream_plasticity	Replay-driven learning potential.	Controls counterfactual exploration.
belief_update_rate	Adaptive synaptic and inferential updating.	Modulates future candidate revision.

11. Dream branch, replay and controlled self-evolution

The dream branch is Melampo's controlled mechanism for counterfactual replay, rare-case exploration, contradiction rehearsal, and mismatch-resolution learning. The enterprise plan specifies that synthetic traces are learning candidates, not medical facts, and that promotion requires rational-control validation and provenance [M3]. This is a critical safety constraint.

The biological analogy is memory replay during sleep and offline consolidation. Biological organisms do not only learn during direct perception; they replay, reorganize, and integrate memory. Melampo adopts the architectural idea without claiming biological equivalence. During low-activity states, the dream branch can generate variations of cases, symptoms, imaging findings, epidemiological contexts, and differential alternatives. These synthetic variations must be tested against rational controls before being inserted into memory as candidates.

Dream-generated material must never silently contaminate real clinical evidence. It must carry source type, synthetic provenance, generation parameters, validation status, and reviewer status. A synthetic path can be useful for robustness, curriculum learning, rare-case simulation, and model critique, but only when the system can distinguish it from patient-derived or literature-derived evidence.

12. Diagnostic orchestration and final authority

The final diagnostic authority inside the research scaffold belongs to MelampoDiagnosticOrchestrator [M1-M2]. This does not mean autonomous clinical authority. It means that within the software architecture, no external model is allowed to produce the final research diagnostic result by itself. The orchestrator integrates pipeline output, differential hypotheses, intuition, area dynamics, policy state, critique, dream output, and model capability records.

The final structured result should contain a case identifier, result label, top hypothesis, differential list, intuition summary, Melampo metrics, support by modality, policy, critique, dream alternatives, model capability decision record, and audit trace. If the policy requires abstention, if mismatch is excessive, if uncertainty is high, or if risk thresholds are violated, the orchestrator must emit abstain_or_escalate rather than a misleading confident label.

Output field	Purpose
result_label	Research label or abstain_or_escalate state.
top_hypothesis	Leading candidate with score and rationale.
differential	Alternative hypotheses with support and contradiction.
intuition	Selected intuition and candidate scores.
melampo_metrics	PI score, prediction error, mismatch, convergence, candidate temperature, belief update.
support	Visual, textual, contextual, epidemiological and semantic evidence.
policy	Allow, abstain, escalate, thresholds and observed values.
critique	Warnings, suggestions and priority review.
dream	Counterfactual alternatives and auto-evolution plan.
audit_trace	Trace of final authority, limitations and pipeline reasoning.

13. Validation, calibration and prospective study design

Melampo separates validation into retrospective benchmarking, prospective validation, and calibration [M4]. This separation is essential because high retrospective accuracy does not guarantee prospective clinical utility; good discrimination does not imply well-calibrated risk; and a calibrated score can still be unsafe under distribution shift. Calibration has been described as a central weakness of predictive analytics when ignored, because risk estimates can mislead even when discrimination appears adequate [S15-S16].

The benchmark layer uses labeled, de-identified datasets with gold labels, slices, provenance, and dataset cards. It reports sample count, answered count, abstention count, top-1 accuracy, coverage, selective accuracy, and slice metrics. The prospective validation layer locks a prediction before outcome is known, stores a payload hash and timestamp, and attaches outcome only later. The calibration layer computes expected calibration error, maximum calibration error, Brier score, calibration bins, and threshold suggestions.

Figure 4. Validation ladder for clinical AI research governance

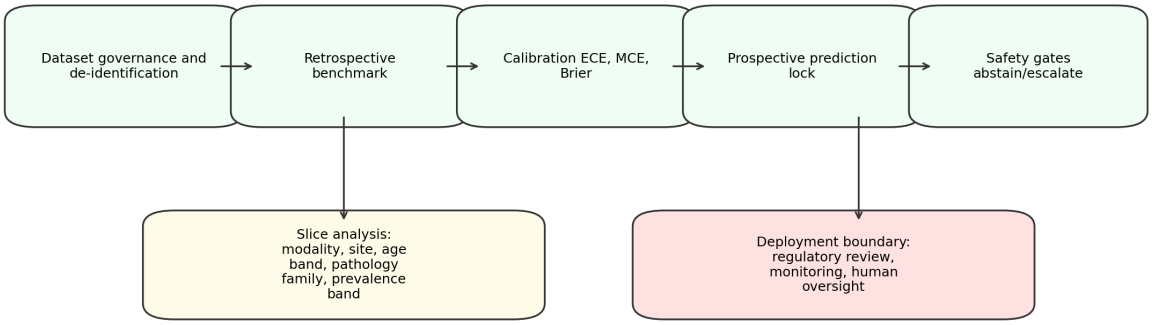

Figure 4. Validation ladder. Retrospective benchmarking, calibration, prospective prediction locking, and safety gates are distinct requirements.

13.1 Benchmark JSONL format

```
{"case_id":"case-001","payload":{"case_id":"case-
001"},"gold_labels":["pneumonia"],"slices":{"modality":"XR","site":"demo"},"provenance":{"dataset"
:"demo"}}
```

13.2 Calibration interpretation

- Expected Calibration Error estimates average confidence-accuracy disagreement across bins.
- Maximum Calibration Error captures the worst bin-level gap.
- Brier score measures squared error between probability and correctness.
- Calibration must be sliced by modality, pathology family, site, scanner/protocol, prevalence band, learning status, and source type.
- Calibration is insufficient without prospective validation and clinical oversight.

14. Operational instructions

14.1 Local development setup

```
python -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -e .[dev]
pytest -q
melampo-decision-record
melampo-weaviate-schema
```

14.2 Enterprise optional setup

```
python -m pip install -e .[enterprise]
```

```
melampo-weaviate-schema --output schema.json
```

```
melampo-weaviate-schema --execute --endpoint https://WEAVIATE_ENDPOINT.example --api-key-env
WEAVIATE_API_KEY
```

The --execute path is intentionally conservative. The core adapter prepares schema materialization and requires infrastructure-specific subclassing for live Weaviate calls. This preserves the no-hidden-network-call rule.

14.3 Minimum operational governance checklist

- Create dataset cards before running benchmarks.
- Verify de-identification and access class before ingesting clinical data.
- Record source provenance for every document and memory object.
- Run retrospective benchmark before prospective validation.
- Run calibration before adjusting thresholds.
- Lock prospective predictions before outcome attachment.
- Keep dream-generated records in candidate status until rational-control validation and human review.
- Keep external models behind explicit configuration and audit logs.
- Never allow an external model to bypass MelampoDiagnosticOrchestrator.

15. Risk, governance and regulatory boundary

The main technical risks are hallucination, retrieval error, ontology mismatch, hidden distribution shift, poor calibration, overconfident imaging interpretation, synthetic data contamination, prompt injection through documents, provenance loss, and ambiguous final authority. The main clinical risks are false reassurance, delayed escalation, missed rare disease, misclassification of urgent pathology, and inappropriate generalization across sites or populations.

The governance response is architectural. Melampo requires abstention, escalation, provenance, validation slices, calibration, prospective prediction locking, no hidden network calls, candidate status for synthetic memory, and final authority in a transparent orchestrator. These constraints do not eliminate risk, but they make risk visible and auditable.

Risk	Mitigation in Melampo
Hallucinated reasoning	Ground by Weaviate objects and provenance; use critique loop and abstention.
Image overconfidence	Pillar-0 remains signal provider; calibrate and cross-check with text/context.
Retrieval mismatch	Use object-property relations, source metadata and ontology references.
Synthetic contamination	Default dream records to candidate; require validation before promotion.
Poor calibration	Compute ECE, MCE and Brier; slice by modality/site/pathology.
Distribution shift	Prospective validation and monitoring before deployment.
Opaque final decision	Use MelampoDiagnosticOrchestrator and audit trace.

16. Research roadmap

9. Implement infrastructure-specific Weaviate subclass for live schema materialization, upsert and search.
10. Implement a Pillar-0 backend adapter for CT/MRI volumetric signal extraction and visual area signal generation.

11. Implement a Gemma 4 backend adapter for grounded clinical text and agentic workflow reasoning.
12. Implement an optional Claude-style critic adapter behind the critique loop.
13. Implement governed Docling source catalogs and document-to-Weaviate ingestion pipelines.
14. Define dataset-specific benchmark loaders and dataset cards for NIH ChestX-ray14, Open-i, LIDC-IDRI, and institutionally approved internal datasets when available.
15. Run retrospective benchmark suites by modality, pathology family, and site.
16. Calibrate candidate confidence, abstention thresholds, and orchestrator policy thresholds.
17. Establish prospective validation with immutable prediction registry and outcome adjudication.
18. Develop regulatory and clinical governance package before any deployment beyond research.

17. Formal conclusion

Project Melampo proposes a principled architecture for computer-aided clinical intuition. Its distinctive contribution is not the selection of a single frontier model, nor the construction of another RAG application. Its contribution is the integration of specialized perception, grounded language reasoning, semantic object memory, neuro-dynamic coherence metrics, dream replay, differential reasoning, critique, policy, calibration, and final orchestration into a single auditable research scaffold.

The system is medically ambitious but appropriately bounded. Its architecture recognizes that medicine requires context, contradiction, time, uncertainty, provenance, and humility. For that reason, Melampo assigns final internal authority to an orchestrator rather than to a model, requires abstention and escalation, separates synthetic learning from clinical truth, and embeds validation into the research path. The project should be advanced as a scientific instrument for studying multimodal diagnostic reasoning and computer-aided intuition, not as an autonomous diagnostic device.

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Appendix A. Minimal JSON structures

A.1 Retrospective benchmark record


```
{"case_id":"case-001","payload":{"case_id":"case-
001"},"gold_labels":["pneumonia"],"slices":{"modality":"XR","site":"demo"},"provenance":{"dataset"
:"demo"}}
```


A.2 Prospective validation event


```
{"prediction_id":"...","case_id":"...","payload_hash":"...","created_at":...,"diagnostic_result":{
...},"locked":true,"outcome":null}
```


A.3 Weaviate semantic object outline


```
{"class":"ClinicalCase","properties":{"case_id":"...","learning_status":"candidate","provenance":{
}},"references":{"hasSymptom":[],"hasImage":[],"hasDifferential":[]},"named_vectors":{"case_summar
y_vector":[],"case_trace_vector":[]}}
```

Appendix B. Site re-ingestion requirement

The public site melampo.chandra.it should be re-ingested in a future revision using a reliable crawler or manual export. Its contents should be compared against the GitHub documentation corpus and incorporated only where provenance, authorship, version, and publication status are clear.

Appendix C. Formal computational specification

C.1 Functional-area signal contract

A Melampo functional area should emit a structured area signal rather than a free-form statement. The minimal contract is: area identity, model provider identity where applicable, signal list, signal count, salience score, uncertainty score, focus, provenance, limitations, ontology references, and relation candidates. This contract allows downstream modules to compare visual, textual, contextual, and epidemiological signals without assuming that every model exposes the same native interface.

The signal contract is intentionally conservative. It treats every external model output as evidence under uncertainty. A model may be highly capable, but its contribution remains a hypothesis-bearing signal until it has been grounded, cross-checked, calibrated, and accepted by the final orchestrator. This design prevents architectural privilege from being granted to a single provider because of fluency, novelty, or benchmark performance.

Field	Required meaning	Example
area	Functional source of the signal.	visual_diagnostic, language_listening, epidemiology.
salience_score	Estimated importance or intensity of the signal.	0.0 to 1.0.
uncertainty_score	Estimated unreliability or incompleteness.	0.0 to 1.0.
ontology_refs	Clinical concept references.	SNOMED-like, ICD-like, RadLex-like.
relations	Candidate graph links.	Symptom suggests Pathology; Finding supports Pathology.
provenance	Source and processing trace.	model version, document page, dataset, timestamp.

C.2 Candidate scoring grammar

A candidate diagnostic hypothesis H should be described by a tuple: H = (label, domain, support, contradiction, uncertainty, severity, actionability, provenance). The candidate is never only a label. The support component should include multimodal evidence; contradiction should include explicit mismatch; uncertainty should include missing evidence and calibration limits; severity and actionability should inform escalation even when likelihood is uncertain.

A simplified research scoring grammar can be expressed as: candidate_score = f(support, convergence, prior_context, semantic_grounding, deductive_gate) - g(contradiction, prediction_error, mismatch_index, uncertainty, bias_risk). This grammar is not a validated clinical formula. It is an engineering constraint: positive evidence and negative evidence must remain separately inspectable.

Appendix D. Physiological analogy and limitations

D.1 Why the analogy must remain bounded

Melampo uses physiological and neuroscientific language as a source of design constraints, not as a literal biological claim. Terms such as prediction error, inhibition, salience, dream replay, and belief update describe computational analogues. They do not imply that the software reproduces cortical columns, thalamic loops, neuromodulatory nuclei, synaptic plasticity, or conscious medical understanding.

The strongest scientific use of the analogy is architectural. Biological cognition separates perception, memory, attention, action selection, error correction, and learning. Medical reasoning separates observation, hypothesis formation, differential diagnosis, risk assessment, and action. Melampo uses this separation to avoid monolithic model authority. The analogy becomes unsafe only if it is interpreted as proof of biological fidelity or clinical validity.

D.2 Interoception, context and diagnosis

Clinical diagnosis is not limited to external observation. Symptoms include subjective and interoceptive states: pain, dyspnea, fatigue, dizziness, palpitations, nausea, anxiety, and malaise. A diagnostic system must therefore treat patient narrative as a first-class signal rather than a secondary caption for imaging. The language_listening_area exists to preserve this narrative signal and to prevent visual findings from dominating when the clinical story contradicts them.

Interoceptive symptoms are often noisy, culturally mediated, temporally variable, and hard to calibrate. They require temporal structure. Acute chest pain, chronic exertional dyspnea, recurrent fever, progressive weakness, and sudden neurological deficit imply different latent dynamics. A future Melampo implementation should therefore enrich the language area with temporal parsing, symptom trajectory modeling, and links to pathophysiological mechanisms.

Appendix E. Pathophysiological mapping requirements

A mature Melampo memory layer should not only connect symptoms to diseases. It should connect symptoms, findings, and risks to mechanisms. For example, fever can be linked to inflammatory cytokine signaling, infection, malignancy, autoimmune disease, drug reaction, or thromboinflammatory processes. Dyspnea can be linked to ventilation-perfusion mismatch, impaired diffusion, cardiac pump failure, anemia, neuromuscular weakness, or anxiety-mediated hyperventilation. A semantic graph that stores only labels will not support expert-level differential reasoning.

The object-property schema should therefore evolve toward mechanism nodes: inflammatory process, ischemic process, hemorrhagic process, obstructive process, neoplastic process, infectious process, metabolic process, autoimmune process, degenerative process, toxic process, and functional process. These mechanism nodes should not replace pathologies; they should provide explanatory bridges between symptoms, imaging findings, laboratory abnormalities, and differential hypotheses.

Mechanism node	Example signals	Diagnostic function
Inflammatory process	Fever, high CRP, tissue edema.	Links infection, autoimmune disease, malignancy and tissue injury.
Ischemic process	Sudden pain, infarct imaging, elevated lactate.	Groups vascular occlusion and perfusion failure.
Hemorrhagic process	Hyperdensity on CT, anemia, shock.	Supports urgent escalation and imaging review.
Obstructive process	Dilation upstream, colic, airflow limitation.	Connects anatomy with mechanical pathophysiology.
Neoplastic process	Mass effect, weight loss, nodal spread.	Links imaging, systemic symptoms and staging logic.
Metabolic process	Electrolyte disturbance, encephalopathy.	Protects against purely anatomical reasoning.

Appendix F. Implementation protocol for real datasets

F.1 Dataset onboarding

19. Create a dataset card before ingestion. The card must identify source, access class, license, population, modality, inclusion criteria, exclusion criteria, label process, adjudication method, known biases, and intended evaluation use.
20. Verify de-identification and legal access. No personally identifiable data should enter the repository. Research data should remain in controlled storage.
21. Map labels to normalized concepts. Raw labels should be preserved, but benchmark labels should be normalized to a stable concept vocabulary.
22. Define slices before evaluation. At minimum: modality, site, age band, sex or gender where lawful, pathology family, prevalence band, scanner or protocol, source type, and learning status.
23. Freeze the benchmark manifest before model evaluation. Post-hoc slice discovery should be marked exploratory.
24. Run retrospective benchmark and calibration before any prospective protocol.
25. Pre-register prospective thresholds, abstention behavior, escalation criteria, and stopping rules.

F.2 Prospective validation protocol

A prospective validation protocol must lock the Melampo output before the outcome is known. The payload hash, prediction timestamp, model capability registry version, schema version, calibration version, and policy thresholds should be recorded. Outcome adjudication should be independent from the model output where possible. Any later modification of the prediction record should produce a new audit event rather than overwrite the original record.

Appendix G. Regulatory and deployment maturity levels

Level	Name	Required evidence
0	Conceptual research	Architecture, documentation, toy examples, no clinical claims.
1	Research scaffold	Executable pipeline, tests, model cards, dataset cards, validation primitives.
2	Retrospective evaluation	De-identified benchmark datasets, calibration, slice analysis.
3	Prospective silent validation	Locked predictions, real workflow, outcomes attached later, no clinical action.
4	Human-in-the-loop research support	Ethics approval, specialist review, monitoring, incident process.
5	Regulated clinical deployment	Regulatory clearance, quality system, post-market surveillance.

Melampo currently occupies the research scaffold level with elements supporting retrospective and prospective evaluation. It should not be described as a clinical diagnostic product. The correct near-term objective is a robust research platform capable of generating auditable evidence for or against future clinical development.

Supplement I. Extended technical-scientific elaboration

This supplement expands the theoretical and operational core of the treatise. It is included to clarify the relationship between physiology, clinical reasoning, enterprise architecture, semantic memory, validation, and future translational research. The supplement does not change the safety boundary: Melampo remains a research scaffold and not a validated clinical device.

I.1 Multimodal perception as a clinical necessity

Clinical reasoning is fundamentally multimodal. A physician does not interpret a radiological image in isolation from the patient narrative, and does not interpret the narrative without considering epidemiology, physiology, medication exposure, prior disease, and the expected natural history of candidate conditions. Melampo formalizes this clinical reality by maintaining separate functional areas rather than prematurely flattening all evidence into a single embedding. Visual information, text, context, and epidemiology retain their own salience and uncertainty before being integrated.

In radiology, the distinction is especially important because an imaging abnormality can be incidental, chronic, technically induced, clinically irrelevant, or clinically decisive depending on context. A small pulmonary nodule, a subtle intracranial hyperdensity, bowel wall thickening, ground-glass opacity, or ventricular enlargement does not carry a fixed meaning outside patient age, symptoms, time course, prior imaging, comorbidities, and pre-test probability. A computer-aided intuition system must therefore preserve the link between visual features and clinical context.

The multimodal design also protects against a common failure mode of modern AI systems: the collapse of evidence into a persuasive explanation. Language generation can hide uncertainty behind coherence. Visual classification can hide uncertainty behind a probability score. Retrieval can hide uncertainty behind similarity. Melampo counteracts this by forcing each modality to expose its contribution as a structured, auditable signal.

I.2 Neurophysiology of signal integration and its computational analogue

Biological neural systems integrate signals across multiple time scales. Fast ion-channel events generate action potentials; synaptic integration shapes local circuit behavior; recurrent networks maintain state; neuromodulatory systems adjust gain, salience, arousal, and learning; hippocampal and cortical circuits consolidate memory across longer intervals. Melampo does not simulate these processes mechanistically, but it borrows the principle that reliable cognition emerges from regulated interaction among signal generation, gain control, memory, and error correction.

The computational analogue is organized around salience, uncertainty, prediction error, convergence, and inhibition. Salience represents signal strength or relevance; uncertainty represents unreliability or incompleteness; prediction error represents mismatch between expected and observed evidence; convergence represents multi-area agreement; inhibition represents the suppression of overdominant or weakly grounded candidates. This vocabulary allows the system to avoid treating confidence as a single number divorced from evidence structure.

In clinical terms, this matters because strong findings can be wrong and weak findings can be decisive. A subtle imaging sign may dominate if it explains a dangerous pathology and matches the time course. A strong textual claim in a report may be discounted if it conflicts with imaging, epidemiology, or objective measurement. Melampo is designed to represent this kind of flexible signal regulation explicitly.

I.3 Predictive processing and differential diagnosis

Predictive processing provides a useful formal analogy for differential diagnosis. A diagnostic hypothesis predicts a set of expected observations. Pneumonia predicts compatible symptoms, imaging patterns, inflammatory markers, risk factors, and time course. Pulmonary embolism predicts a different constellation. Heart failure, malignancy, autoimmune disease, drug toxicity, and functional syndromes predict still different configurations. The differential process can therefore be seen as competition among generative hypotheses that attempt to explain available evidence with minimum residual contradiction.

Melampo operationalizes this through prediction error and mismatch metrics. When a candidate explains visual and textual evidence but fails epidemiological or contextual constraints, the mismatch index should remain high. When multiple functional areas converge around the same candidate and contradiction remains low, the convergence index and deductive gate can increase. When evidence is incomplete or contradictory, the candidate

temperature should rise, allowing additional alternatives to remain active.

This approach is clinically conservative. It favors structured uncertainty over premature certainty. The goal is not to force every case into a diagnostic label, but to produce a controlled state: leading candidate, alternatives, support, contradiction, missing evidence, uncertainty, and policy. In ambiguous states, abstention or escalation is a successful outcome, not a failure.

I.4 Pathophysiology-aware semantic memory

A mature clinical memory should represent more than symptom-disease edges. It should represent mechanisms. Dyspnea can arise from airway obstruction, parenchymal disease, vascular obstruction, cardiac dysfunction, anemia, neuromuscular weakness, metabolic acidosis, anxiety, or mixed states. Chest pain can arise from ischemic, inflammatory, musculoskeletal, pulmonary, esophageal, vascular, or psychophysiological mechanisms. Fever can arise from infection, malignancy, autoimmune disease, drug reaction, thromboinflammatory disease, or tissue necrosis.

Pathophysiology-aware memory allows Melampo to reason across surface forms. Two diseases may share symptoms because they share mechanisms. Two imaging findings may support different diagnoses because they indicate a shared process. A semantic graph that includes mechanisms can therefore support analogical reasoning, rare-case recognition, and contradiction detection. For example, a candidate diagnosis that explains fever but fails to explain focal neurological deficit should remain incomplete unless a bridging mechanism is found.

This memory should be linked to ontology codes but not reduced to codes. SNOMED-like, ICD-like, RadLex-like, and LOINC-like identifiers provide normalization and interoperability. However, clinical reasoning requires explanatory relations: causes, supports, contradicts, increases risk of, belongs to case, found in study, temporally precedes, and requires exclusion. Melampo should evolve toward this richer semantic relation set.

I.5 The physiology of uncertainty and clinical abstention

Uncertainty is not a defect of clinical reasoning; it is a physiological and epistemic condition of diagnosis. Biological systems operate under uncertainty because sensory information is partial, noisy, delayed, and context-dependent. Clinicians operate under uncertainty because patients present early, atypically, incompletely, or with multiple comorbidities. A safe medical AI system must not hide uncertainty. It must expose it, quantify it where possible, and act conservatively when uncertainty exceeds safe bounds.

Melampo separates uncertainty into several operational forms: missing evidence, contradiction, low grounding, high mismatch, poor calibration, distribution shift, and unsafe policy state. These forms should not be collapsed into a single confidence score. A candidate may be well grounded but still uncertain because a dangerous alternative remains unexcluded. Another candidate may have high visual support but poor textual or epidemiological support. Another may be plausible but outside the calibration domain.

Abstention is therefore a first-class clinical behavior. In ordinary classification benchmarks, abstention may appear as non-response. In medicine, abstention can be the correct action when the system lacks sufficient evidence. Melampo should distinguish between benign abstention, urgent escalation, request for missing data, and specialist referral.

I.6 Enterprise RAG as clinical memory governance

Retrieval-augmented generation in clinical AI must be governed differently from general-purpose document search. The retrieved unit should not be merely a passage of text. It should be a governed evidence object with source, date, license, section, page, ontology references, relation type, confidence, and scope. A guideline recommendation, a case memory, a synthetic dream trace, and a radiology report excerpt have different epistemic status even if their text embeddings are close.

Weaviate is appropriate for Melampo because the object-property structure maps naturally onto this requirement. Clinical objects can carry both vectors and properties. Named vectors can support distinct representations for report text, image patterns, case summaries, document layout, and ontology context. Cross-references can represent clinical relations. This architecture makes it possible to retrieve not only similar text but also connected meaning.

The RAG layer should also defend against contamination. Literature, local protocol, de-identified case trace, and dream-generated counterfactual should remain separate source classes. A synthetic trace can be useful for training or robustness testing, but it must not be retrieved as clinical evidence without explicit labeling. Governance metadata is therefore part of the data model, not an optional annotation.

I.7 Dream replay as constrained generative exploration

The dream branch is a generative exploration mechanism. Its purpose is to create counterfactual variants, rare-case rehearsals, contradiction scenarios, and mismatch-resolution hypotheses. It should operate primarily in low-activity or offline research settings, never as uncontrolled real-time mutation of clinical memory. Its outputs should be clearly marked as synthetic and should enter memory only as candidate material.

The clinical value of dream replay lies in robustness. Diagnostic systems fail when they see only canonical cases. Rare disease, mixed pathology, atypical presentation, and partial evidence require exploration beyond the common pattern. A dream branch can generate structured variants that test whether the reasoning system can maintain differential breadth, detect contradiction, and avoid premature closure.

The danger of dream replay is epistemic contamination. If synthetic material becomes indistinguishable from real clinical evidence, the system can train itself into fiction. Melampo therefore requires rational-control validation, provenance, learning status, and human review before promotion. The dream branch is a hypothesis generator, not a source of truth.

I.8 Calibration as the bridge between score and action

A diagnostic score is not clinically meaningful unless its relationship to observed correctness is known. A model that assigns 0.9 confidence should be correct close to 90 percent of the time in the relevant clinical slice, or its score is not calibrated. Calibration matters because clinical action depends on thresholds: whether to reassure, request a test, escalate, monitor, or intervene.

Melampo includes calibration primitives because the same raw score can have different meaning across modalities, diseases, sites, and populations. A confidence score derived from chest radiography cannot be assumed to calibrate on abdominal CT. A model calibrated in one hospital cannot be assumed to calibrate in another. A rare disease class cannot borrow calibration from common classes without evidence.

The calibration layer should therefore produce global metrics and slice metrics. Expected calibration error, maximum calibration error, Brier score, calibration curves, and threshold suggestions are research instruments. They should inform policy thresholds and orchestrator behavior, but never replace prospective validation.

I.9 Prospective validation and clinical translation

Prospective validation is the boundary between retrospective promise and clinical relevance. Retrospective benchmarks can reveal useful performance, but they cannot fully capture workflow timing, missing data, distribution shift, clinician behavior, or outcome adjudication. A prospective protocol must lock predictions before outcomes are known, preserve payload hashes, record model and policy versions, and attach outcomes only after adjudication.

Melampo already includes a file-based prospective validation registry as a research primitive. A clinical-grade implementation should replace it with immutable storage, role-based access control, audit trails, protocol identifiers, and compliance review. The principle is more important than the storage engine: predictions must not be altered after outcome knowledge, and outcome adjudication should not be contaminated by model output when avoidable.

No architecture, however elegant, can substitute for prospective evaluation. The clinical claim of a diagnostic system begins only when it is tested in the intended workflow, on the intended population, with defined endpoints, thresholds, and safety gates. Until that point, Melampo should be described as a research scaffold and scientific instrument.

I.10 The final scientific position

The scientific value of Melampo is the disciplined integration of multiple lines of reasoning: radiological perception,

clinical language understanding, semantic graph memory, physiological analogy, predictive processing, differential diagnosis, controlled self-evolution, calibration, and prospective validation. Each component is insufficient alone. Their interaction is the object of research.

The project should resist two temptations. The first is the temptation to grant authority to the newest external model. The second is the temptation to overstate biological fidelity. Melampo should remain a transparent architecture in which models are replaceable, metrics are inspectable, uncertainty is explicit, and safety boundaries are preserved.

If developed according to these principles, Melampo can become a platform for studying artificial diagnostic intuition: not intuition as mystical guessing, but intuition as an emergent, auditable, multimodal, memory-grounded, uncertainty-aware state transition. That is the scientific identity of the project.